

Break Tank Package Booster Set for Fire
Fighting

HyaSolo 2 D FL Compact

Type Series Booklet



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Type Series Booklet HyaSolo 2 D FL Compact

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Building Services: Water Supply

Break Tank Package Booster Set for Fire Fighting

HyaSolo 2 D FL Compact



Main applications

- Fire-fighting systems to DIN 14462

Fluids handled

- Fire-fighting water
- Liquids not containing aggressive, abrasive or solid substances

Operating data

Table 1: Operating properties

Characteristic		Value
Flow rate	Q [m³/h]	≤ 36
	Q [l/s]	≤ 10
Head	H [m]	≤ 160
Fluid temperature	T [°C]	≥ 0
		≤ +30
Operating pressure	p [bar]	≤ 16
Inlet pressure	p _d [bar]	≤ 8
Power supply	V [V]	3/PE, AC 400, 50 Hz

Design details

Design

- Fully automatic break tank package booster set for fire fighting to EN 1717 and EN 13077 with type AB air gap
- Baseplate-mounted
- Modular design (comprising the following modules):
 - Hydraulic unit with control cabinet (single-pump station incl. temperature sensor, pressure gauge, accumulator, pressure switch and solenoid valve) as well as valves and piping
 - Baseplate

- Break tank unit with inlet device, flushing device, overflow monitoring and dry running monitoring
- Vibration damping

Installation

- Stationary dry installation

Drive

- Three-phase asynchronous motor
- 380 / 420 V, 50 Hz
- 2 poles
- Enclosure IP55
- Efficiency class IE3 to IEC 60034-30
- Soft starter

Automation

- Electro-mechanical control system, IP54
- Manual-automatic selector switch
- External master switch
- Timer for functional check run every 24 hours and hygienic flushing function of the inlet line once a week
- Phase monitoring relay
- In automatic mode, the motor protection circuit only outputs an alert.
- Terminal strip/terminals with identification for all connections
- Pressure switch on the discharge side for starting and stopping, monitored for broken wires and short circuit
- Additional option of starting and stopping by remote ON/OFF, monitored for broken wires and short circuit
- Pressure switch for monitoring pressure build-up
- Pressure-controlled starting and pressure-controlled stopping
- Circuit diagram to VDE and parts list for electric parts

Indication via indicator lamps:

- Colour LEDs indicating lack of water and no pressure build-up (red)
- Colour LED indicating a fault (amber)
- Colour LED indicating readiness for operation (green)
- Colour LED indicating operation started by pressure switch or remote ON/OFF (white)

Indication via volt-free isolating terminals:

- Lack of water
- No pressure build-up
- Overflow
- Position of manual-automatic selector switch
- Pump control via pressure switch or remote ON/OFF
- General fault message
- Drinking water stop

Monitoring for broken wires and monitoring for short circuit:

- > 10.3 kΩ: broken wire
- 10 kΩ: good state
- 1 kΩ: pump start through actuation of limit switch
- < 300 Ω: short circuit

Designation

Example: HyaSolo 2 D FL 1/1508 Compact

Table 2: Designation key

Code	Description
HyaSolo	Type series
2	Generation
D FL	Break tank package booster set for fire fighting with three-phase motor
1	Number of pump sets
15	Pump size
08	Number of stages
Compact	Compact design

Configuration and function



Fig. 1: Illustration of the break tank package booster set for fire fighting

1	High-pressure centrifugal pump	4	Inlet
2	Control cabinet	5	Break tank
3	Baseplate		

Design

Fully automatic, ready-to-connect break tank package booster set for fire fighting comprising one high-pressure centrifugal pump (1), a control cabinet (2) and a break tank (5) for the hygienic separation of drinking water and fire-fighting water. The modular design allows separating the break tank package booster set for fire fighting into three modules (hydraulic unit with control cabinet, tank unit, and baseplate). Anti-vibration mounts are fitted at the pump feet and baseplate (3).

Function

The pump set is started and stopped as a function of pressure via the electro-mechanical control system. If the pressure drops below the set start-up pressure p_E , the pump set is started up via the pressure switch. When consumption decreases again, the pump set is stopped as a function of pressure after the set minimum runtime (up to 10 minutes). Independently of the pressure switch, the break tank package booster set for fire fighting can also be controlled via remote ON/OFF (e.g. a switch in the wall hydrant). The pressure switch and remote ON/OFF input are monitored for broken wires and short circuit. If a broken wire or short circuit are detected, the break tank package booster set for fire fighting is started up automatically and a fault is indicated.

Dry running protection stops the pump sets in automatic mode as soon as no more fluid is pumped (default setting). Tripping of the motor protection switch triggers a fault message.

The required quantity of the fluid handled is supplied via the float valve.

The float valve at the drinking water inlet is opened and closed fully automatically as a function of the level in the break tank. A float switch monitors for lack-of-water and overflow, and

signals such events to the control system. Power supply of the flushing system and water level monitoring equipment is effected via terminals in the control cabinet.

Via a ball valve fitted at the drain connection of the tank, the function of the mechanical inlet valves can be checked once a month.

Materials

Table 3: Overview of available materials

Part No.	Description	Material
Pump		
101	Pump casing	Stainless steel
101	Pump shroud	Stainless steel
655	Hydraulic system	Stainless steel
310	Plain bearing	Aluminium oxide
433	Mechanical seal	To EN 12756
433.01	Primary ring	Silicon carbide
433.02	Mating ring	Carbon graphite
-	Elastomer	EPDM
Hydraulic design		
700	Piping	Chrome steel
79.01	Control unit	Steel, painted
-	Accumulator	Connection made of stainless steel
159-12	Membrane	Approved for drinking water
Shut-off valves		
102	Body	Nodular cast iron
-	Valve disc	Stainless steel
-	Liner material	EPDM-XV
-	Break tank	Polyethylene
742	Swing check valve	Cast steel
174-4	Float valve	Brass / copper / EPDM

Product benefits

- Ready to connect due to standardised dimensions
- Design to DIN 14462 ensures reliable operation in the event of fire.
- Hygienic separation of drinking water and fire-fighting water eliminates any risk of contamination.
- Hygienically safe owing to inlet valve approved for drinking water and automatic flushing
- Straightforward installation due to modular design and drain connection with integrated odour trap
- Operating reliability provided by a pressure switch for starting and stopping, and a pressure switch for monitoring the system pressure
- Easy to install and transport in buildings where access space is restricted as the modular design allows separating the assemblies "hydraulic unit with control cabinet" and "break tank unit", mounted on a common baseplate.

Product information

Product information as per Regulation No. 1907/2006 (REACH)

For information as per European chemicals regulation (EC) No. 1907/2006 (REACH) see <https://www.ksb.com/en-global/company/corporate-responsibility/reach>.

Selection information

Lightning protection

- Electrical installations must be protected against overvoltage (binding since 14 December 2018) (see DIN VDE 0100-443 (IEC 60364-4-44:2007/A1:2015, modified) and DIN VDE 0100-534 (IEC 60364-5-53:2001/A2:2015, modified)). Whenever modifications are made to existing installations, retrofitting a surge protective device (SPD) in accordance with VDE is mandatory.
- A maximum cable length of 10 metres should not be exceeded between the surge protective device (usually type 1, internal lightning protection) installed at the service entrance and the equipment to be protected. For longer cables, additional surge protective devices (type 2) must be provided in the sub-distribution board upstream of the equipment to be protected or directly in the equipment itself.
- The associated lightning protection concept must be provided by the operator or by a suitable provider commissioned by the operator. Surge protective devices can be offered for the control units on request.

Selecting the break tank package booster set for fire fighting

 For selecting the break tank package booster set for fire fighting see KSB EasySelect and/or the Planning Information for Fire-fighting Water Systems (reference number 2300.023).

Example

Specify the operating points:

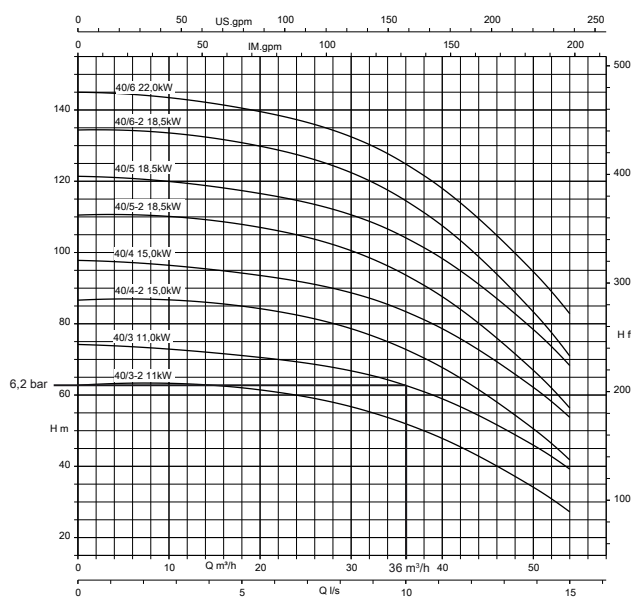
- Flow rate required: 36 m³/h
- Head: 62 m (start-up pressure p_E : 6.2 bar)

Calculation:

- Transfer the values to the selection chart to select the corresponding pump size.

⇒ Movitec 40/3

⇒ HyaSolo 2 D FL 1/4003 Compact



Selection chart for determining the break tank package booster set for fire fighting (example)

Technical data

HyaSolo 2 D FL Compact with Movitec V

Table 4: Technical data

Size	p _E ¹⁾ + Δp ²⁾ = p _A ³⁾						Connection	P _N	I _N	Starting current (soft start) ⁴⁾	Minimum start- ing current	Total rated power	Effective volume	Mat. No.	[kg]
	18 m³/h 5,0 l/s		24 m³/h 6,66 l/s		36 m³/h 10,0 l/s										
	p _E	Δp	p _E	Δp	p _E	Δp			3~400 V, 50 Hz						
	[bar]	[bar]	[bar]	[bar]	[bar]	[bar]			[kW]						
1/1504	4,2	1,3	-	-	-	-	R 2	4,00	7,45	36,7	25,3	5,2	400	29135241	294
1/1505	5,2	1,7	-	-	-	-	R 2	5,50	10,00	51,0	35,2	6,9	400	29135242	339
1/1506	6,3	2,1	-	-	-	-	R 2	7,50	13,40	68,1	47,2	9,3	400	29135243	340
1/1507	7,4	2,4	-	-	-	-	R 2	7,50	13,40	68,1	47,2	9,3	400	29135244	345
1/1508	8,5	2,7	-	-	-	-	R 2	11,00	19,30	89,6	61,8	13,4	400	29135245	404
1/1509	9,5	3,2	-	-	-	-	R 2	11,00	19,30	89,6	61,8	13,4	400	29135246	495
1/1510	10,5	3,6	-	-	-	-	R 2	11,00	19,30	89,6	61,8	13,4	400	29135247	496
1/2503	5,4	0,5	5,0	0,9	-	-	DN 65	5,50	10,00	51,0	35,2	6,9	400	29135248	351
1/2504	7,2	0,8	6,7	1,3	-	-	DN 65	7,50	13,40	68,1	47,2	9,3	400	29135249	358
1/2505	9,0	1,0	8,4	1,6	-	-	DN 65	11,00	19,30	89,6	61,8	13,4	400	29135250	442
1/2506	10,7	1,3	10,0	2,0	-	-	DN 65	11,00	19,30	89,6	61,8	13,4	400	29135251	445
1/2507	12,5	1,5	11,4	2,6	-	-	DN 65	15,00	26,20	121,6	83,8	18,2	400	29135252	459
1/4003-2	-	-	-	-	4,8	1,3	DN 80	11,00	19,30	89,6	61,8	13,4	600	29135253	405
1/4003	-	-	-	-	5,4	1,9	DN 80	11,00	19,30	89,6	61,8	13,4	600	29135254	405
1/4004-2	-	-	8,0	0,5	6,8	1,7	DN 80	15,00	26,20	121,6	83,8	18,2	600	29135255	409
1/4004	-	-	9,2	0,5	8,0	1,7	DN 80	15,00	26,20	121,6	83,8	18,2	600	29135256	423
1/4005-2	-	-	10,5	0,5	9,3	1,7	DN 80	18,50	31,80	147,2	101,8	22,0	600	29135257	428
1/4005	-	-	11,5	0,6	10,0	2,1	DN 80	18,50	31,80	147,2	101,8	22,0	600	29135258	444
1/4006-2	-	-	12,7	0,7	11,2	2,2	DN 80	18,50	31,80	147,2	101,8	22,0	600	29135259	446
1/4006	-	-	13,2	1,3	12,1	2,4	DN 80	22,00	37,60	185,3	127,8	26,1	600	29135260	486

Selection aid for control cabinets

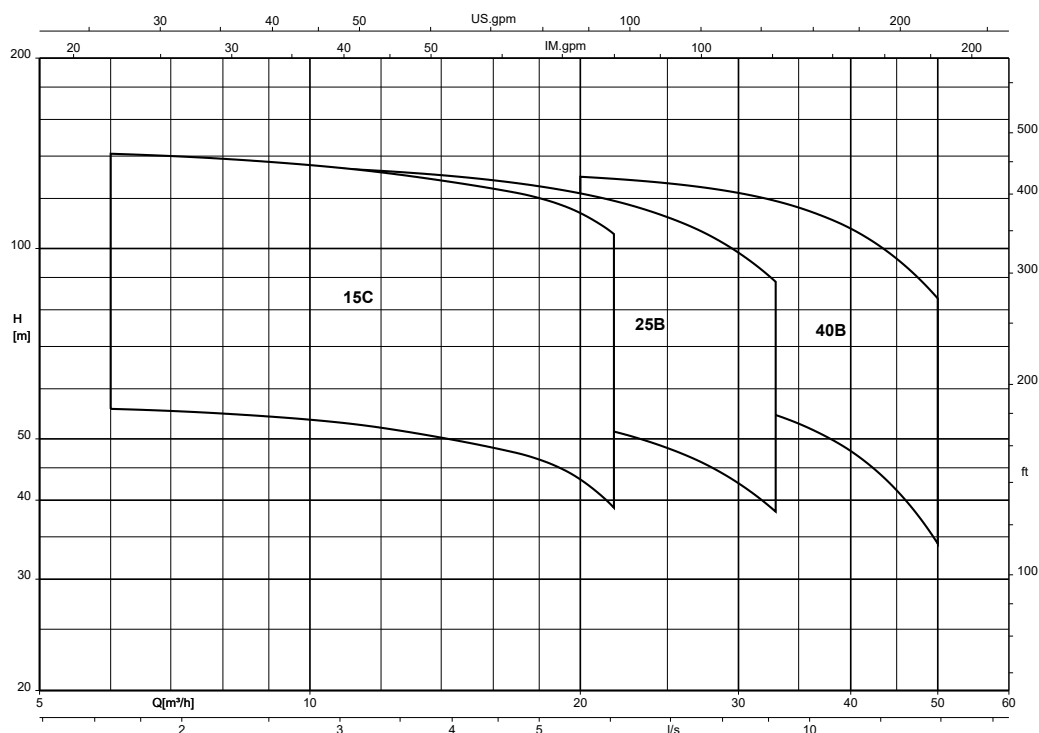
Table 5: Overview of control cabinets

Motor rating per pump	Maximum connection cross-section		Control cabinet size
	Stranded	Solid	H x W x D
[kW]	[mm ²]	[mm ²]	[mm]
5,5	6	10	600 x 800 x 250
7,5	6	10	600 x 800 x 250
11	10	16	600 x 800 x 250
15	16	25	600 x 800 x 250
18,5	16	25	600 x 800 x 250
22	16	25	1000 x 600 x 250
30	35	50	1000 x 600 x 250
37	50	50	1000 x 800 x 300

- ¹ Start-up pressure
- ² Maximum pressure increase
- ³ Stop pressure
- ⁴ Recommended triggering characteristic of fuse elements: gG or C

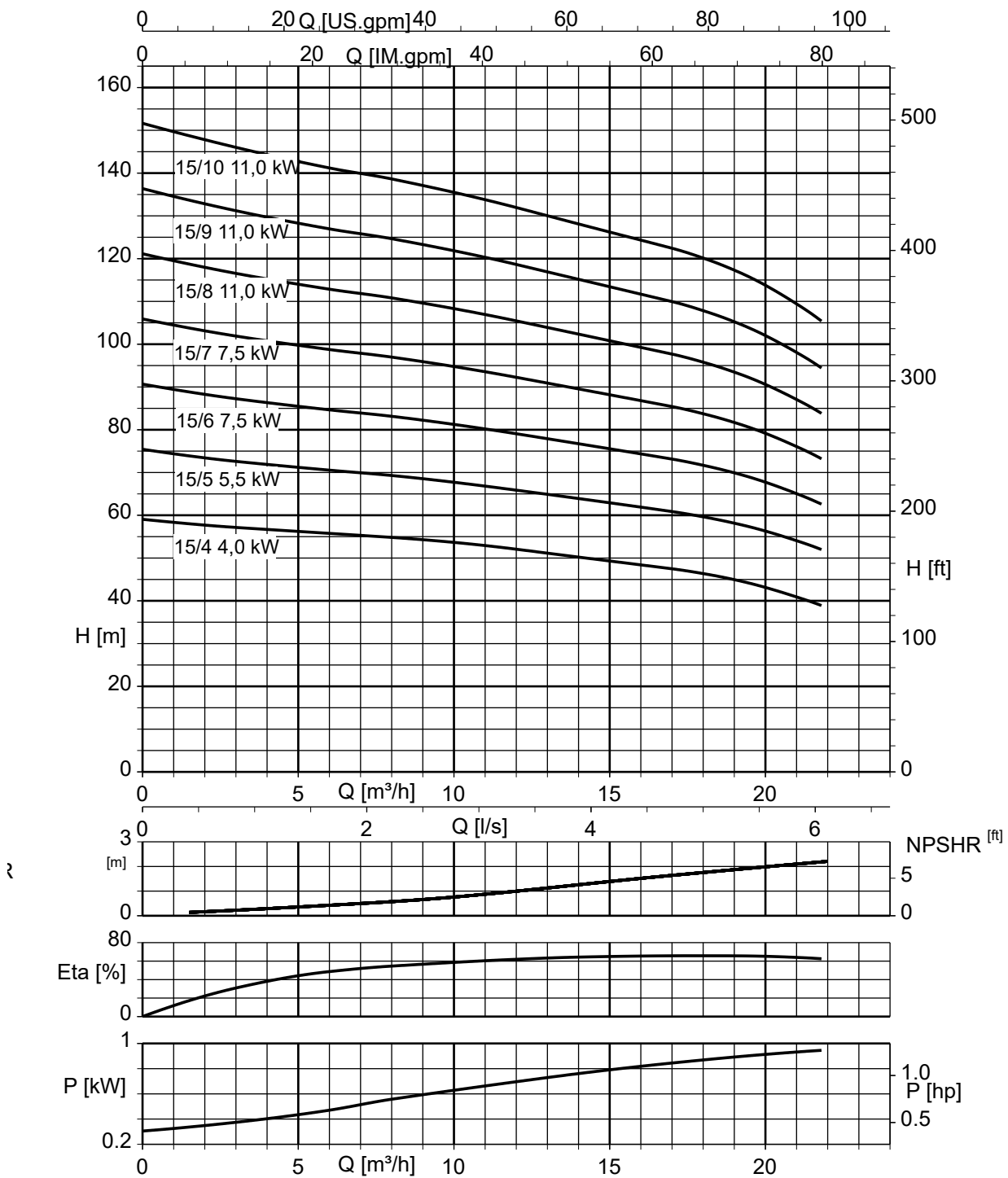
Selection chart

HyaSolo 2 D FL Compact, n = 2900 rpm

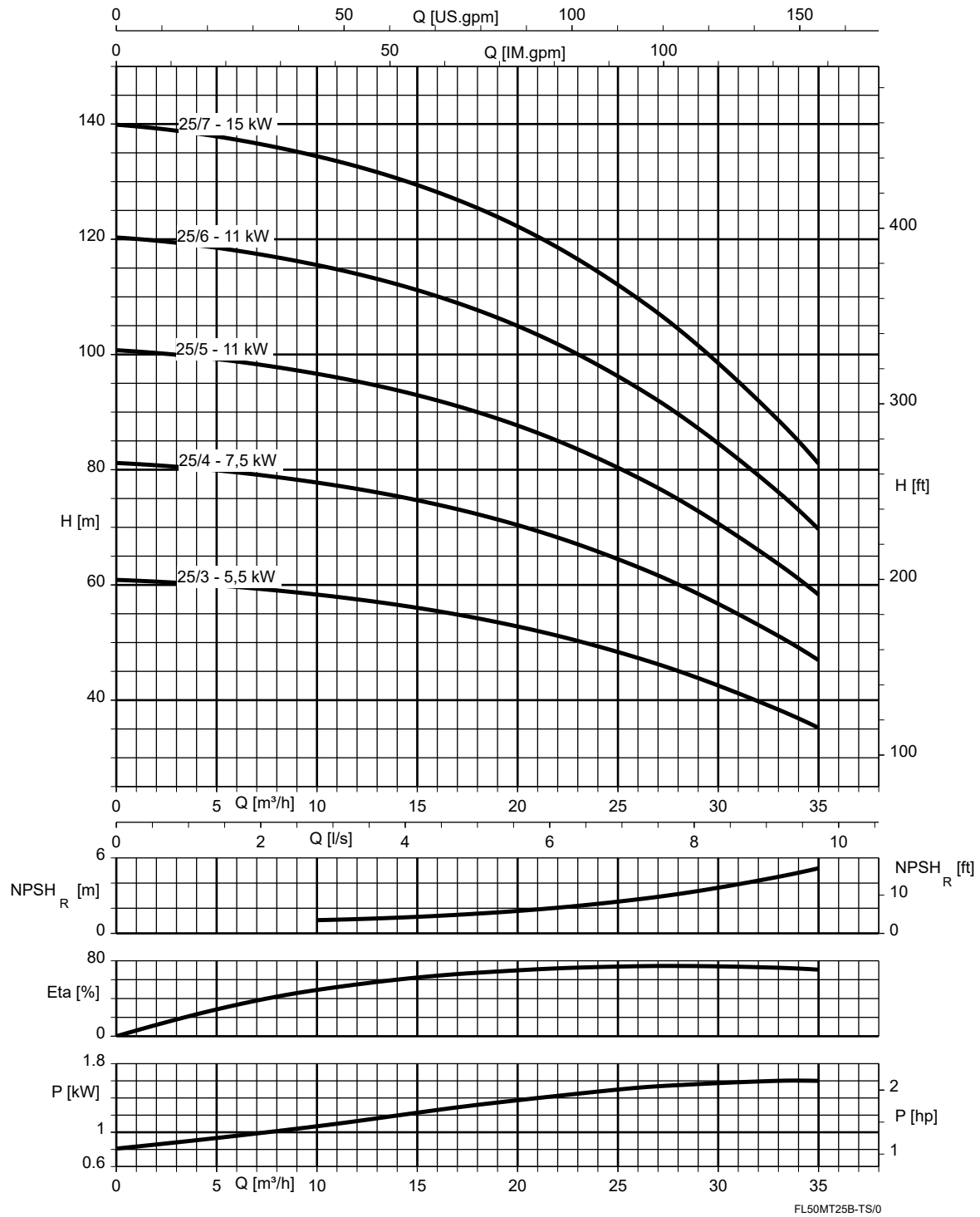


Characteristic curves

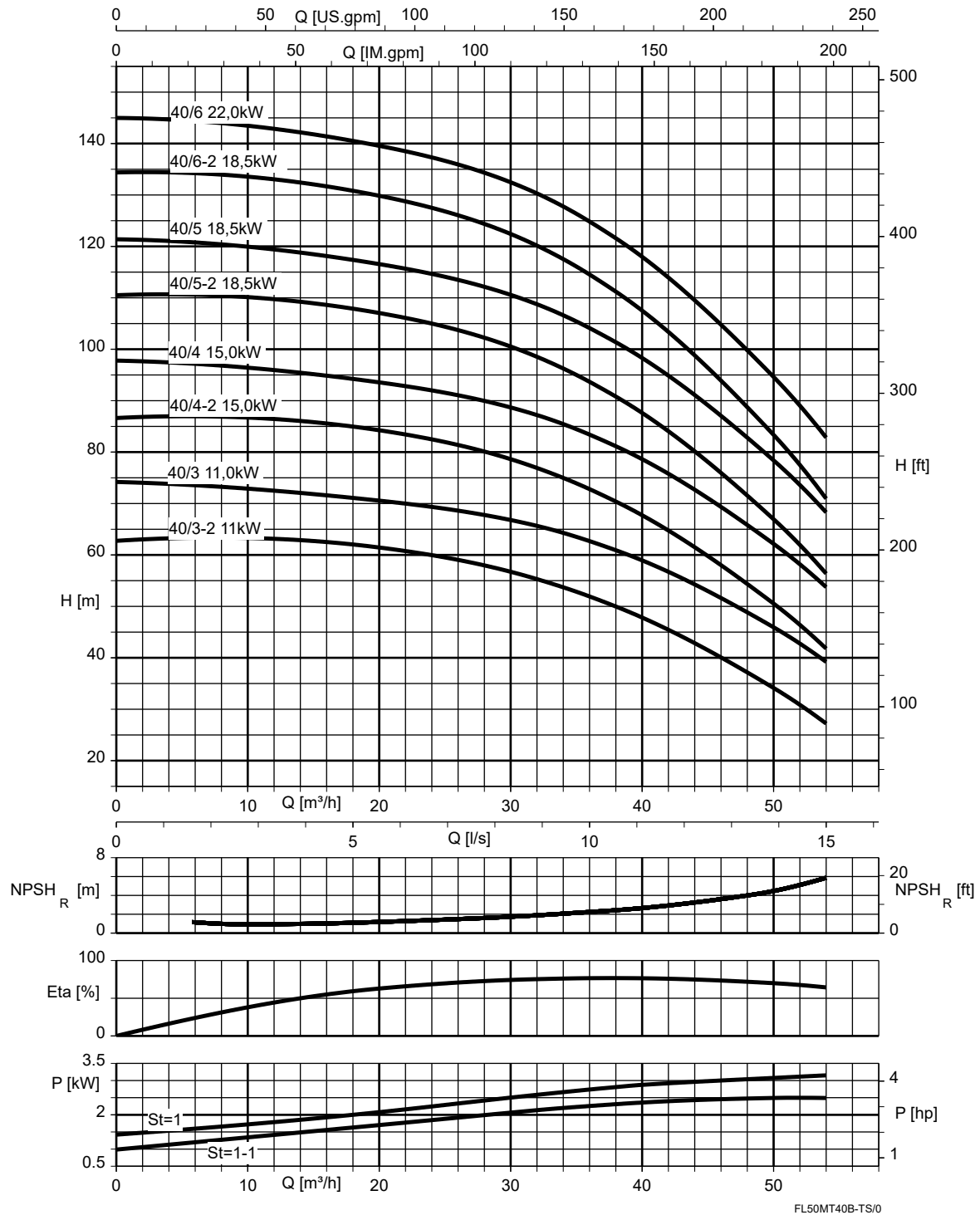
HyaSolo 2 D FL Compact with Movitec 15C, n = 2900 rpm



HyaSolo 2 D FL Compact with Movitec 25B, $n = 2900$ rpm



HyaSolo 2 D FL Compact with Movitec 40B, $n = 2900$ rpm



Dimensions

HyaSolo 2 D FL Compact

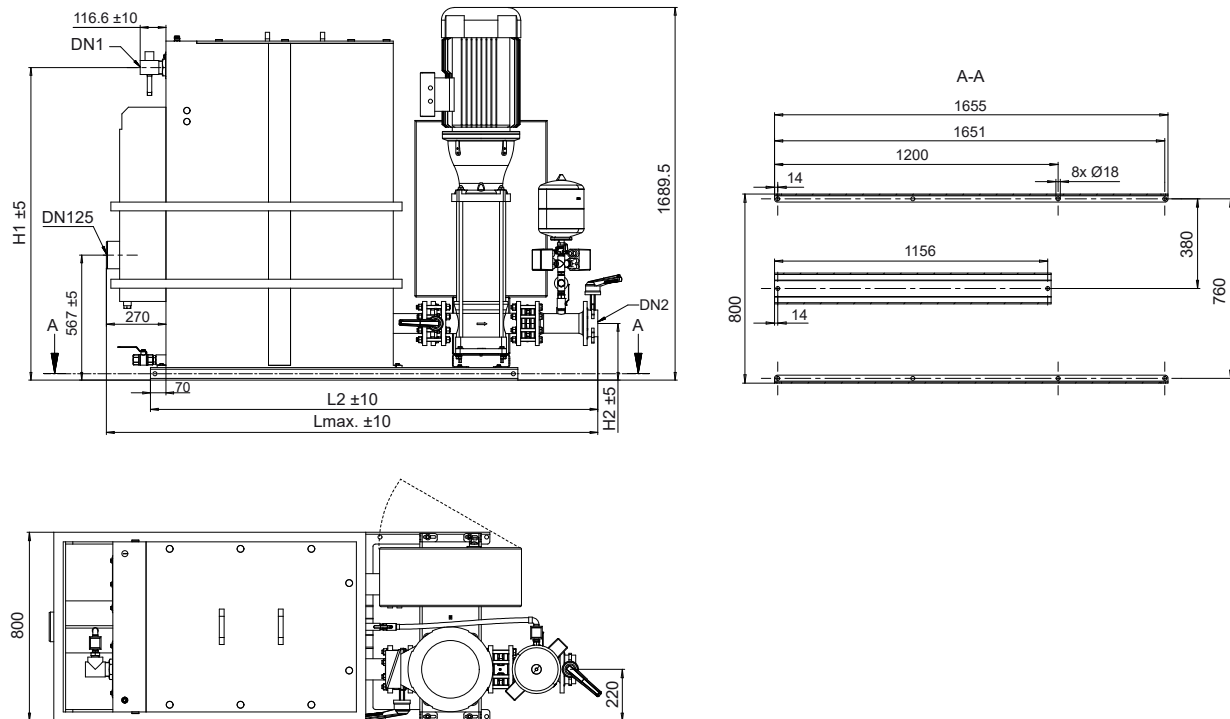


Fig. 2: Dimensions of the break tank package booster set for fire fighting [mm]

 Flanges drilled to EN 1092-1 PN 16

Table 6: Dimensions

Size	Connection		H1	H2	L2	Lmax
	DN 1	DN 2	[mm]	[mm]	[mm]	[mm]
1/15...	G 1 1/4	DN 50	1362	207	2004	2204
1/25...	G 1 1/4	DN 65	1362	222	2020	2220
1/40...	G 2	DN 80	1417	257	2028	2228

Scope of supply

Depending on the model, the following items are included in the scope of supply:

- Break tank package booster set for fire fighting
 - 1 vertical pump set (Movitec high-pressure centrifugal pump incl. IE3 motor)
 - Discharge-side piping
 - Check valve
 - Suction-side and discharge-side shut-off valves
 - Discharge-side, direct-flow membrane-type accumulator
 - Pressure switch on the discharge side for starting and stopping
 - Pressure switch for monitoring pressure build-up
 - Base frame
 - Temperature sensor
 - Automatic flushing device
 - Break tank with a type AB air gap to EN 1717
 - Drain connection at container for collecting overflow water with integrated odour trap
 - Sensors for lack-of-water monitoring and overflow monitoring
 - Fill level indication
 - Float valve for mains water supply to DVGW
- Control unit
 - Power supply connection
 - Motor protection switch
 - External master switch
 - Broken wire monitoring relay and short-circuit monitoring relay
 - Manual-automatic selector switch
 - Timer
 - Phase monitoring relay
 - Soft starter
 - Control transformer
 - Terminal strip with markings for all connections
 - Plug-type connection for the control cabinet and break tank

Accessories

 See the separate type series booklet Accessories for Pressure Booster Systems 1954.5.



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